BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Mathematics (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** 5191

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Order of transpose of [2 1; 0 1; 3 2] is:

(A) 3-by-2
(B) 2-by-3
(C) 1-by-3
(D) 3-by-1

قالب [2 1; 0 1; 3 2] کے ٹرانسپوز کا آرڈر ہے:

Q2. Product of [xy] [2; -1] is equal to:

(A) [2x + y]
(B) [x - 2y]
(C) [2x - y]
(D) [x + 2y]

ضرب حاصل [xy] [2; -1] برابر ہے:

Q3. In $\sqrt[3]{35}$ the radicand is:

(A) 3
(B) 1/3
(C) 35
(D) $(35)^3$

$\sqrt[3]{35}$ میں ریڈی کینڈ ہے:

Q4. Real part of $2ab(i + i^2)$ is:

(A) 2ab
(B) -2ab
(C) 2abi
(D) 2abi

$2ab(i+i^2)$ کا حقیقی حصہ ہے:

Q5. The logarithm of unity to any base is:

(A) 1
(B) 10
(C) e
(D) 0

کسی اساس پر '1' کا لوگارتھم برابر ہوتا ہے:

Q6. The value of log(p/q) is:

(A) log p - log q
(B) log p / log q
(C) log p + log q
(D) log q - log p

$\log(p/q)$ کی قیمت ہے:

Q7. 4x + 3y - 2 is an algebraic:

(A) Expression
(B) Sentence
(C) Equation
(D) Inequation

$4x+3y-2$ ایک الجبریک _____ ہے۔

(A) جملہ
(B) فقرہ
(C) مساوات
(D) غیر مساوات

Q8. Factors of $8x^3 + 27y^3$ are:

(A) $(2x + 3y)(4x^2 + 9y^2)$
(B) $(2x - 3y)(4x^2 - 9y^2)$
(C) $(2x + 3y)(4x^2 - 6xy + 9y^2)$
(D) $(2x - 3y)(4x^2 + 6xy + 9y^2)$

$8x^3+27y^3$ کے اجزائے ضربی ہیں:

Q9. H.C.F. of $p^3q - pq^3$ and $p^5q^2 - p^2q^5$ is:

(A) $pq(p^2 - q^2)$
(B) $pq(p - q)$
(C) $p^2q^2(p - q)$
(D) $pq(p^3 - q^3)$

جملوں p^3q-pq^3 اور $p^5q^2-p^2q^5$ کا عاد اعظم ہے:

Q10. The square root of $a^2 - 2a + 1$ is:

(A) $\pm(a + 1)$ (B) $\pm(a - 1)$

(C) $(a - 1)$ (D) $(a + 1)$

$a^2 - 2a + 1$ کا جزر المربع ہے:

Q11. If x is no larger than 10, then:

(A) $x \geq 8$ (B) $x \leq 10$

(C) $x > 10$ (D) $x < 10$

اگر x کی قیمت 10 سے زیادہ نہ ہو تو:

Q12. If (x, 0) = (0, y), then (x, y) is equal to:

(A) (0, 1) (B) (1, 0)

(C) (0, 0) (D) (1, 1)

اگر $(0, y) = (x, 0)$ ہو تو (x, y) برابر ہے:

Q13. Distance between the points (1, 0) and (0, 1) is:

(A) 0 (B) 1

(C) √2 (D) 2

نقاط $(0, 1)$ اور $(1, 0)$ کا درمیانی فاصلہ ہے:

Q14. The _____ altitudes of an isosceles triangle are congruent.

(A) 2 (B) 3

(C) 4 (D) 5

متساوی الساقین مثلث کے _____ ارتفاعات متشابہ ہوتے ہیں۔

Q15. If two medians of a triangle are congruent then the triangle will be:

(A) Isosceles (B) Equilateral

(C) Right angled (D) Acute angled

اگر ایک مثلث کے دو وسطے متشابہ ہوں تو وہ مثلث _____ ہوگی۔

(A) متساوی الساقین **(B)** مساوی الاضلاع

(C) قائمۃ الزاویہ **(D)** حادۃ الزاویہ

Mathematics (Subjective) Group - I

Time Allowed: 02:10 Hours | **Maximum Marks:** 60 | **Paper Code:** 5191

> Note: Part - I (حصہ اول)

Q2. (Marks 12) Attempt any SIX parts:

(i) Find the values of a, b, c and d which satisfy the matrix equation: [a+c a+2b; c-1 4d-6] = [0 -7; 3 2d]

(ii) Multiply these matrices: [2 3; 1 1; 0 -2] [2 -1; 3 0]

(iii) Simplify: $5^{23} \div (5^2)^3$

(iv) Simplify: $\sqrt{(25x^{10}n^5y^8m)}$

(v) Find the value of x from the given statement: $\log_x 64 = 2$

(vi) Write into sum or difference: log (21×5)/8

(vii) Evaluate $(x^2y^3 - 5z^4)/(xyz)$ for x = 4, y = -2, z = -1

(viii) Factorize: $3x - 243x^3$

(ix) Reduce the rational expression in lowest form: $((x+y)^2 - 4xy)/(x-y)^2$

Q3. (Marks 12) Attempt any SIX parts:

(i) Use factorization to find the square root: $x^2 + 1 + 1/(4x^2)$ $(x \neq 0)$

(ii) Solve for x: $|(x+5)/(2-x)| = 6$

(iii) Define a linear inequality in one variable.

(iv) Verify whether the point (0, 0) lies on the line 2x - y + 1 = 0 or not?

(v) Draw the graph of: y = 7

(vi) The end point P of a line segment PQ is (-3, 6) and its mid-point is (5, 8). Find coordinates of the end point Q.

(vii) Define scalene triangle.

(viii) If ΔABC ≅ ΔLMN, then m∠M ≅ _____, m∠N ≅ _____ [triangles with ∠C=60°, ∠A=90°, ∠B=30° and ∠M=30°, ∠L=90°, ∠N=60°]

(ix) The given figure LMNP is a parallelogram and sum of the opposite angles of it is 110°. Find the remaining angles. [parallelogram with ∠N=55°]

Q4. (Marks 12) Attempt any SIX parts:

(i) If the given triangle ABC is equilateral and AD is bisector of angle A, then find the values of unknowns x°, y° and z°. [equilateral triangle with altitude AD]

(ii) 3cm, 4cm and 7cm are not the lengths of triangle. Give the reason.

(iii) Define ratio.

(iv) Verify that the triangle having the following measures of sides are right-angled: a = 5cm, b = 12cm, c = 13cm

(v) Find the value of x in the given figure: [right triangle with hypotenuse 13cm, base 5cm, height x]

(vi) Define the term "area of a figure".

(vii) Find the area of the given figure: [rectangle 6cm × 3cm]

(viii) Define incenter.

(ix) Construct a triangle XYZ: mZX = 6.4cm, mYZ = 2.4cm, m∠Y = 90°

> Note: Part - II, Attempt any THREE questions. Each question carries 08 marks. Question No. 9 is compulsory.

> حصہ دوم، کوئی سے تین سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 08 نمبر ہیں۔ سوال نمبر 9 لازمی ہے۔

Q5(a). (Marks 04) Solve the system of linear equation by using the matrix inverse method: $4x + 2y = 8$; $3x - y = -1$

Q5(b). (Marks 04) Solve the equation for real x and y: $(2 - 3i)(x + iy) = 4 + i$

Q6(a). (Marks 04) Solve by the logarithm: $(8.97)^3 \times (3.95)^2 / \sqrt[3]{15.37}$

Q6(b). (Marks 04) If $p = 2 + \sqrt{3}$ then find $p^2 - 1/p^2$.

Q7(a). (Marks 04) Factorize: $(x + 2)(x + 3)(x + 4)(x + 5) - 15$

Q7(b). (Marks 04) Use division method to find the square root of the expression: $x^4 - 10x^3 + 37x^2 - 60x + 36$

Q8(a). (Marks 04) Solve the equation: $5(x - 3)/6 - x = 1 - x/9$

Q8(b). (Marks 04) Construct the ΔABC and draw perpendicular bisectors of its sides: mBC = 2.9cm, m∠B = 60°, m∠A = 30°

Q9. (Marks 08) Prove that any point on the right bisector of a line segment is equidistant from its end points. OR Prove that any point on the bisector of an angle is equidistant from its arms.

GROUP – II

Mathematics (Objective) Group - II

Time Allowed: 20 Minutes	**Maximum Marks:** 15	**Paper Code:** 5196

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Factors of $27x^3 - 1/x^3$ are:

(A) $(3x - 1/x)(9x^2 + 3 + 1/x^2)$

(B) $(3x + 1/x)(9x^2 + 3 + 1/x^2)$

(C) $(3x - 1/x)(9x^2 - 3 + 1/x^2)$

(D) $(3x + 1/x)(9x^2 - 3 + 1/x^2)$

$27x^3 - 1/x^3$ کے اجزائے ضربی ہیں:

Q2. Conjugate of surd $a + \sqrt{b}$ is:

(A) $-a + \sqrt{b}$

(B) $a - \sqrt{b}$

(C) $a + \sqrt{b}$

(D) $\sqrt{a} - \sqrt{b}$

مقدار $a + \sqrt{b}$ کا مزدوج جملہ ہے:

Q3. $\log(m^n)$ can also be written as:

(A) $(\log m)^n$

(B) m log n

(C) n log m

(D) log emn

$\log(m^n)$ کو _____ بھی لکھا جا سکتا ہے:

Q4. log e _____ where e ≈ 2.718 :

(A) 1

(B) 0.4343

(C) ∞

(D) 1

$\log e$ = _____ جبکہ $e \approx 2.718$:

Q5. Real part of $2ab(i + i^2)$ is:

(A) 2ab

(B) -2ab

(C) 2abi

(D) -2abi

کمپلیکس نمبر $2ab(i + i^2)$ کا حقیقی حصہ ہے:

Q6. In $\sqrt[3]{35}$ the radicand is:

(A) 3

(B) 1/3

(C) 35

(D) $(35)^3$

$\sqrt[3]{35}$ میں ریڈی کینڈ ہے:

Q7. If X + [1 0; 0 -1] = [1 0; 0 1], then X is equal to:

(A) [2 2; 2 0]
(B) [0 2; 2 2]
(C) [2 0; 0 2]
(D) [2 2; 0 2]

اگر $X + [1\ 0; 0\ -1] = [1\ 0; 0\ 1]$ ہو تو X برابر ہے:

Q8. [√2 0; 0 √2] is called _____ matrix.

(A) Zero
(B) Unit
(C) Scalar
(D) Singular

$[\sqrt{2}\ 0; 0\ \sqrt{2}]$ کو _____ قالب کہا جاتا ہے۔

(A) صفری
(B) وحدانی
(C) سکیلر
(D) سنگولر

Q9. An angle on the base of an isosceles triangle is 30°. What is the measure of its vertical angle?

(A) 30°
(B) 60°
(C) 90°
(D) 120°

متساوی الساقین مثلث کے قاعدے پر ایک زاویہ 30° ہے۔ اس کے راسی زاویے کی مقدار کیا ہے؟

Q10. The right bisectors of the three sides of a triangle are:

(A) Congruent
(B) Collinear
(C) Concurrent
(D) Parallel

مثلث کے تینوں اضلاع کے عمودی ناصف ہوتے ہیں:

(A) متشابہ
(B) ہم خط
(C) ہم نقط
(D) متوازی

Q11. Mid-point of the points (-2, 2) and (2, -2) is:

(A) (2, 2)
(B) (-2, -2)
(C) (0, 0)
(D) (1, 1)

نقاط (2, -2) اور (-2, 2) کا درمیانی نقطہ ہے:

Q12. Which ordered pair satisfies the equation y = 2x?

(A) (1, 2)
(B) (2, 1)
(C) (2, 2)
(D) (0, 1)

کون سا نقطہ مساوات $y = 2x$ کے گراف پر واقع ہے؟

Q13. A statement involving any of the symbols <, >, ≤ or ≥ is called:

(A) Equation
(B) Identity
(C) Inequality
(D) (0, 1)

کوئی بیان جس میں <, >, ≥ یا ≤ میں سے کوئی ایک علامت پائی جائے کہلاتی ہے:

(A) مساوات
(B) مماثلت
(C) غیر مساوات

Q14. The square root of $x^4 + 1/x^4 + 2$ is:

(A) $\pm(x + 1/x)$
(B) $\pm(x^2 + 1/x^2)$
(C) $\pm(x - 1/x)$
(D) $(x^2 + 1/x^2)$

$x^4 + 1/x^4 + 2$ کا جزر المربع ہے:

Q15. H.C.F. of $5x^2y^2$ and $20x^3y^3$ is:

(A) $5x^2y^2$
(B) $20x^3y^3$
(C) $100x^5y^5$
(D) $5xy$

جملوں $5x^2y^2$ اور $20x^3y^3$ کا عاد اعظم ہے:

Mathematics (Subjective) Group - II

Time Allowed: 02:10 Hours | **Maximum Marks:** 60 | **Paper Code:** 5196

Note: Part - I (حصہ اول)

Q2. (Marks 12) Attempt any SIX parts:

(i) If B = [1 1; 2 0], then verify that $(B^t)^t = B$.

(ii) If B = [1; -1], then find B + [-2; 3].

(iii) Express (-7 + 3i)(-3 + 2i) in the standard form a + bi.

(iv) Evaluate: i^{50}

(v) Express 5.06×10^{10} in ordinary notation.

(vi) Find the value of x when $\log_x 64 = 2$

(vii) Reduce the expression to the lowest form: $(x^2 - 4x + 4)/(2x^2 - 8)$

(viii) Simplify: $(3 + \sqrt{3})(3 - \sqrt{3})$

(ix) Factorize: $9xy - 12x^2y + 18y^2$

Q3. (Marks 12) Attempt any SIX parts:

(i) Use factorization to find the square root of the expression: $4x^2 - 12xy + 9y^2$

(ii) Solve the equation: $(2/3)x - (1/2)x = x + 1/6$

(iii) Solve: $|(3 - 5x)/4| - 1/3 = 2/3$

(iv) Draw the graph of $x = -3$.

(v) Verify that whether the point (5, 3) lies on the line $2x - y + 1 = 0$?

(vi) Find the distance between the given points: A(-8, 1), B(6, 1)

(vii) Find the mid-point of the given points: A(0, 0), B(0, -5)

(viii) If $\Delta ABC \cong \Delta LMN$, then find x: [triangles with $\angle C=60°$, $\angle A=40°$, $\angle B=80°$ and $\angle M=80°$, $\angle L=40°$, $\angle N=x°$]

(ix) If LMNP is a parallelogram then find m, n: [parallelogram with PN=8, $\angle N=5n°$, LM=8m-4n, $\angle L=5m°$]

Q4. (Marks 12) Attempt any SIX parts:

(i) If the given triangle ABC is equilateral triangle and AD is bisector of angle A, then find the values of unknowns x°, y° and z°. [equilateral triangle with altitude AD]

(ii) If 10cm, 6cm and 8cm are the lengths of a triangle, then verify that sum of measures of two sides of a triangle is greater than the third side.

(iii) Define ratio.

(iv) The three sides of a triangle are of measure 8, x and 17 respectively. For what value of x will it become base of a right angled triangle?

(v) Find the value of x in the given figure: [right triangle with hypotenuse 13cm, base 5cm, height x]

(vi) Find the area of the given figure: [parallelogram with base 8cm and height 4cm]

(vii) Define altitude or height of a triangle.

(viii) Construct triangle ABC in which: mAB = 2.5cm, $m\angle A = 30°$, $m\angle B = 105°$

(ix) Define orthocenter.

> Note: Part - II, Attempt any THREE questions. Each question carries 08 marks. Question No. 9 is compulsory.

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Q5(a). (Marks 04) Solve the given system of linear equations by using matrix inversion method: $4x + 2y = 8$; $3x - y = -1$

Q5(b). (Marks 04) Simplify: $(81)^n \times 3^5 - (3)^{4n-1}(243) / (9^{2n})(3^3)$

Q6(a). (Marks 04) Use log table to find the value: 0.8176×13.64

Q6(b). (Marks 04) If $m + n + p = 10$ and $mn + np + mp = 27$, then find the value of $m^2 + n^2 + p^2$.

Q7(a). (Marks 04) For what value of m is the polynomial $p(x) = 4x^3 - 7x^2 + 6x - 3m$ exactly divisible by $x + 2$?

Q7(b). (Marks 04) Find square root by division method: $4x^2 + 12xy + 9y^2 + 16x + 24y + 16$

Q8(a). (Marks 04) Solve the equation: $2/(x^2 - 1) - 1/(x + 1) = 1/(x + 1)$, $x \neq \pm 1$

Q8(b). (Marks 04) Construct triangle PQR and draw its altitudes: mRP = 3.6cm, $m\angle Q = 30°$, $m\angle P = 105°$

Q9. (Marks 08) Prove that any point on the right bisector of a line segment is equidistant from its end points. -- OR -- Prove that any point on the bisector of an angle is equidistant from its arms.